

ENEL445 Roy Smith Optimisation Block

Learning Guide: Least Squares, Convex Optimisation, LMIs, and MPC

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1 Source Materials

本学习指导基于本章 Roy Smith 相关课件、练习和 6 节转录内容整理。主线不是逐字复述 lecture，而是按考试复习逻辑重构：

- Least-Squares Problems and Methods/Lecture 6 Least squares regression part 1.pdf
- Least-Squares Problems and Methods/Lecture 7 Least squares regression part 2.pdf
- Convex Optimisation, LMIs, and MPC/Lecture-8-Convex-optimisation-part-1.pdf
- Convex Optimisation, LMIs, and MPC/Lecture-9-Convex-optimisation-part-2.pdf
- Convex Optimisation, LMIs, and MPC/ Lecture-10- Convex- optimisation- part-3- model-predictive-control-(MPC).pdf
- exercises/LS_exercises.pdf
- exercises/Statistics_exercises.pdf
- exercises/Convex_Opt_exercises.pdf
- transcribe/20260504_Lecture_6_least_squares_regression_part_1.md
- transcribe/20260505_Lecture_7_least_squares_statistics_BLUE.md
- transcribe/20260506_Least_squares_regularisation_validation.md
- transcribe/20260511_Lecture_8_convex_sets_cones_ellipsoid_method.md
- transcribe/20260512_Lecture_9_LMI_SDP_control_design.md
- transcribe/20260513_Lecture_10_model_predictive_control_MPC.md

2 Study Strategy

这组材料是混合型章节：前半部分是 least squares 的计算与统计解释，后半部分是 convex optimisation、LMI 和 MPC 的建模框架。复习时不要按日期机械背，而要按下面的问题链组织：

真实工程问题

-> 数据/系统模型

-> 变量、约束、目标函数

-> 是否是 least-squares 或 convex problem

-> 可用公式、solver 或 iterative method

-> 考试题型和易错点

最重要的三类能力：

1. 会把问题写成 optimisation form。
2. 会解释公式背后的统计或几何意义。
3. 会判断一个问题属于 LS、LP、QP、SOCP、SDP/LMI 或 MPC 中哪一类。

复习优先级：

Priority	Topics
Must know	LS formulation, normal equations, BLUE, bias/variance/MSE, regularised LS, convexity definition, LP/QP/SOCP/SDP/LMI classification, MPC formulation
Should know	nonlinear LS linearisation, ellipsoid method idea, Schur complement, stability/state-feedback LMI ideas, explicit MPC
Recognise only	S-procedure details in convex exercise problem 3, deep implementation details of CVX/YALMIP

3 Mindmap-based Learning Path

学习顺序建议如下：

1. **Least squares as optimisation:** 先理解 $y = \Phi\theta + \epsilon$ 、residual、normal equations。
2. **Least squares as statistics:** 再理解 $\hat{\theta}$ 是随机变量，所以要看 bias、variance、MSE。
3. **Regularisation:** 之后看为什么最小 training residual 不等于最好 prediction。
4. **Convex foundations:** 学 convex set、cone、PSD cone、ellipsoid。
5. **Problem classes:** 把 LP、QP、SOCP、QCQP、SDP/LMI 放在同一张分类图里。
6. **LMI tools:** 用 Schur complement 和 PSD cone 理解 matrix constraints。
7. **MPC:** 把前面 optimisation framework 用到 control: finite horizon、constraints、replanning。

4 Least Squares: Core Formulation

4.1 What Problem Does LS Solve?

Least squares 解决的是：模型无法完全通过所有数据点时，如何选择参数使 residual 最小。

线性模型写成：

$$y = \Phi\theta + \epsilon$$

其中：

- $y \in \mathbb{R}^N$ 是观测输出；
- $\Phi \in \mathbb{R}^{N \times d}$ 是 regressor/design matrix；
- $\theta \in \mathbb{R}^d$ 是待估参数；
- ϵ 是误差或 residual。

标准 LS 问题:

$$\min_{\theta} \|y - \Phi\theta\|_2^2$$

等价 constrained form:

$$\begin{aligned} \min_{\theta, \epsilon} \quad & \|\epsilon\|_2^2 \\ \text{subject to} \quad & y - \Phi\theta = \epsilon \end{aligned}$$

考试中常见任务:

- 给出数据和模型, 写出 Φ 、 y 、 θ ;
- 写出 LS optimisation problem;
- 推出 normal equations;
- 判断能不能使用 closed-form inverse。

4.2 Normal Equations

令

$$J(\theta) = \|y - \Phi\theta\|_2^2$$

对 θ 求导并设为 0:

$$\Phi^T \Phi \theta = \Phi^T y$$

如果 $\Phi^T \Phi$ 可逆:

$$\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T y$$

注意:

- normal equations 总是 consistent;
- 但 inverse form 需要 Φ full column rank;
- 如果 $\Phi^T \Phi$ condition number 很大, 解会对噪声敏感。

4.3 Geometric Interpretation

LS 的几何意义是：把 y 投影到 Φ 的 column space 上。残差

$$r = y - \Phi\hat{\theta}$$

与 column space 正交，因此：

$$\Phi^T r = 0$$

也就是：

$$\Phi^T (y - \Phi\hat{\theta}) = 0$$

这正是 normal equations。

记忆方式：**LS** 不是让每个点都正确，而是让误差向量无法再沿模型空间方向减少。

5 Nonlinear Least Squares and Gauss/Ceres

Lecture 6 用 Gauss 预测 Ceres 轨道说明：least squares 不只是 line fitting，也可以处理 nonlinear model。

非线性模型：

$$y_{\text{obs}} = H(\theta, t) + \epsilon$$

目标：

$$\min_{\theta} \frac{1}{2} \epsilon^T \epsilon$$

其中：

$$\epsilon = y_{\text{obs}} - H(\theta, t)$$

非线性 LS 的基本做法：

1. 从 nominal parameter θ_i 开始。
2. 对 $H(\theta)$ 做 Taylor linearisation：

$$H(\theta) \approx H(\theta_i) + H_{\theta}(\theta - \theta_i)$$

3. 定义:

$$\delta y = y_{\text{obs}} - H(\theta_i), \quad \delta \theta = \theta - \theta_i$$

4. 解线性 LS:

$$\min_{\delta \theta} \|\delta y - H_{\theta} \delta \theta\|_2^2$$

5. normal equations:

$$H_{\theta}^T H_{\theta} \delta \theta = H_{\theta}^T \delta y$$

6. 更新:

$$\theta_{i+1} = \theta_i + \delta \theta$$

考试理解重点:

- 三个观测点给 Gauss 一个 nominal orbit, 但完整 22 个观测点用于 LS refinement;
- nonlinear LS 每一步只是 local quadratic/linear approximation;
- 不能把一次 linearised solution 当成 global exact solution。

6 Statistical Properties of Least Squares

6.1 Why LS Estimate Is Random

如果

$$y = \Phi \theta_0 + \epsilon$$

而 ϵ 是随机噪声, 那么

$$\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T y$$

也是随机变量。

因此考试会关心:

- $\hat{\theta}$ 是否 unbiased;
- variance/covariance 有多大;
- MSE 是否最小;

- correlated noise 时如何调整。

6.2 Bias, Variance, and MSE

Bias:

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta_0$$

Variance:

$$\text{Var}(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2]$$

Mean square error:

$$\text{MSE} = E[(\hat{\theta} - \theta_0)^2]$$

核心关系:

$$\text{MSE} = \text{bias}^2 + \text{variance}$$

易错点: unbiased 不等于 minimum MSE。Unbiased 只是 bias 为 0, 但 MSE 还包括 variance。

6.3 BLUE

BLUE = Best Linear Unbiased Estimator。

如果噪声 zero mean 且 covariance 为 R :

$$E[\epsilon\epsilon^T] = R$$

则 BLUE:

$$\hat{\theta}_{\text{BLUE}} = (\Phi^T R^{-1} \Phi)^{-1} \Phi^T R^{-1} y$$

其 covariance:

$$\text{Cov}(\hat{\theta}_{\text{BLUE}}) = (\Phi^T R^{-1} \Phi)^{-1}$$

当 $R = \sigma^2 I$ 时, BLUE 退化为普通 LS。

考试题型：

- 证明 $Z^T \Phi = I$ 是 linear estimator $\hat{\theta} = Z^T y$ unbiased 的条件；
- 推导 $\text{Cov}(\hat{\theta}) = Z^T R Z$ ；
- 写出 BLUE；
- 解释为什么 low variance measurement 权重更大。

7 Regularisation and Validation

7.1 Why Regularisation Is Needed

普通 LS 给出 best linear unbiased estimate，但未必最小化 MSE。Regularisation 的作用：

- 降低 variance；
- 改善 ill-conditioned problem；
- 用一点 bias 换更好的 prediction。

Regularised LS:

$$\min_{\theta} \|y - \Phi\theta\|_2^2 + \eta\|\theta\|_2^2$$

解为：

$$\hat{\theta} = (\Phi^T \Phi + \eta I)^{-1} \Phi^T y$$

当 $\eta = 0$ ，回到 ordinary LS。随着 η 增大：

- solution 被 shrink；
- variance 下降；
- bias 上升；
- conditioning 变好。

7.2 Bias-variance Tradeoff

Regularisation 的核心不是让 training error 最小，而是让 expected prediction error 更好。

记忆方式：

```
eta small -> flexible estimate, low bias, high variance
eta large -> shrunk estimate, higher bias, lower variance
best eta -> balance point for MSE/prediction
```


7.3 Validation

选择 η 时不能只看 fitting data residual。正确思路：

1. 把 data 分成 fitting/training set 和 validation set。
2. 对不同 η 用 fitting set 拟合。
3. 用 validation set 测试 prediction error。
4. 选择 validation error 最小的 η 。

考试可能问：

- 为什么 validation data 不能参与 fitting；
- 为什么 best training fit 不一定 best prediction；
- regularisation 如何连接到 minimum MSE。

8 Convex Sets and Convex Optimisation

8.1 Convex Set

集合 C convex, 如果：

$$x_1, x_2 \in C, \quad \gamma \in [0, 1] \quad \Rightarrow \quad \gamma x_1 + (1 - \gamma)x_2 \in C$$

直觉：任意两点连线都在集合里。

重要例子：

- hyperplane: $\{x \mid a^T x = b\}$;
- halfspace: $\{x \mid a^T x \leq b\}$;
- norm ball: $\{x \mid \|x - x_c\| \leq r\}$;
- polyhedron: linear inequalities/equalities 的交集；
- ellipsoid:

$$\{x \mid (x - x_c)^T P^{-1}(x - x_c) \leq 1\}$$

重要性质：convex sets 的 intersection 仍然 convex。

8.2 Cones and PSD Cone

Cone:

$$x \in C, \gamma \geq 0 \Rightarrow \gamma x \in C$$

Convex cone 还要求 conic combination 也在集合内。

Positive semidefinite cone:

$$S_+^n = \{X \in S^n \mid X \succeq 0\}$$

PSD 的等价理解:

$$z^T X z \geq 0 \quad \text{for all } z$$

或所有 eigenvalues 非负。

易错点:

- positive semidefinite: eigenvalues ≥ 0 ;
- positive definite: eigenvalues > 0 ;
- cone 不自动 convex。

8.3 Convex Optimisation Problem

标准形式:

$$\begin{aligned} \min_x \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b \end{aligned}$$

其中:

- f_0 和 f_i convex;
- equality constraints 必须 affine。

为什么重要: convex problem 的 local information 可以支持 global solution, solver 更可靠。

9 Problem Classes: LP, QP, SOCP, QCQP, SDP

9.1 Linear Program

LP:

$$\begin{aligned}
& \min_x && c^T x + d \\
& \text{subject to} && Gx \leq h \\
& && Ax = b
\end{aligned}$$

特征:

- linear cost;
- linear constraints;
- feasible region 是 polytope/polyhedron。

例子: diet problem、logistics、Chebyshev center。

9.2 Quadratic Program

QP:

$$\begin{aligned}
& \min_x && \frac{1}{2}x^T Px + c^T x + d \\
& \text{subject to} && Gx \leq h \\
& && Ax = b
\end{aligned}$$

特征:

- quadratic cost;
- linear constraints。

例子:

- constrained least squares;
- portfolio optimisation;
- simple MPC。

9.3 Second-order Cone Problem

SOCP 典型 constraint:

$$\|A_i x + b_i\|_2 \leq c_i^T x + d_i$$

特征:

- linear objective 常见;
- feasible region 是 second-order cones 的 intersection;
- QP 有时可以 reformulate 成 SOCP。

9.4 QCQP

QCQP:

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} \quad & \frac{1}{2}x^T P_i x + q_i^T x + r_i \leq 0 \end{aligned}$$

如果 P_i 满足相应 convexity 条件, feasible region 可以理解为 ellipsoids 的 intersection。

9.5 SDP and LMI

Semidefinite program 的核心是 matrix-valued affine constraint:

$$x_1 F_1 + \cdots + x_N F_N + G \preceq 0$$

这就是 Linear Matrix Inequality。

考试分类技巧:

```
linear cost + linear constraints -> LP
quadratic cost + linear constraints -> QP
norm cone constraints -> SOCP
quadratic constraints -> QCQP
PSD matrix constraints -> SDP/LMI
finite horizon dynamic optimisation -> MPC, often QP
```

10 LMI and Schur Complement

10.1 What Is an LMI?

LMI 是这样的约束:

$$F(x) = F_0 + x_1 F_1 + \cdots + x_n F_n \succeq 0$$

其中矩阵对变量是 affine 的。

它重要是因为很多看起来 nonlinear 的 constraint 可以写成 PSD matrix constraint。

10.2 Schur Complement

典型形式：

$$\begin{bmatrix} Q & S \\ S^T & R \end{bmatrix} \succ 0$$

在 $R \succ 0$ 时，等价于：

$$Q - SR^{-1}S^T \succ 0$$

考试中 Schur complement 常用于：

- 把 norm constraint 写成 LMI;
- 把 quadratic inequality 写成 LMI;
- 处理 stability 和 controller synthesis 中的 matrix inequalities。

易错点：使用 Schur complement 时必须检查 block 的 definiteness 条件。

10.3 Stability LMIs

Continuous-time system:

$$\frac{d}{dt}x(t) = Ax(t)$$

stable iff exists $P \succ 0$ such that:

$$A^T P + P A \prec 0$$

Discrete-time system:

$$x_{k+1} = Ax_k$$

stable iff exists $P \succ 0$ such that:

$$A^T P A - P \prec 0$$

State feedback:

$$u_k = Kx_k$$

closed-loop:

$$x_{k+1} = (A + BK)x_k$$

控制设计中常用变量替换，例如 $V = KY$ ，把对 K 和 Lyapunov matrix 的非线性耦合变成 LMI。

11 Model Predictive Control

11.1 Core Idea

MPC 的基本循环：

1. measure or estimate current state;
2. optimise future state/input trajectory over horizon M ;
3. apply only the first input;
4. move one time step and repeat。

一句话记忆：**plan a full future, execute the first move, then replan.**

11.2 Mathematical Formulation

Dynamics:

$$x_{k+1} = Ax_k + Bu_k + Fw_k$$

Finite horizon cost:

$$J(u, x) = \sum_{i=0}^{M-1} (x_i^T Q x_i + u_i^T R u_i) + x_M^T Q_{\text{terminal}} x_M$$

Typical MPC problem:

$$\begin{aligned} \min_{x, u} \quad & J(u, x) \\ \text{subject to} \quad & x_{k+1} = Ax_k + Bu_k + Fw_k \\ & u_{\text{lb}} \leq u_k \leq u_{\text{ub}} \\ & x_k \in X_{\text{operating}} \\ & x_M \in X_{\text{terminal}} \end{aligned}$$

重要理解：

- dynamics 是 constraints;
- decision variables 通常包括 future inputs 和 future states;
- disturbance forecast 可以作为 feedforward information;
- repeated measurement/re-estimation 提供 feedback。

11.3 Hard, Soft, and Terminal Constraints

Hard constraints:

- 必须满足;
- 例子: actuator limits、safety limits。

Soft constraints:

- 想满足, 但允许违反;
- 通过 penalty cost 处理;
- 例子: building temperature comfort band。

Terminal constraints:

- 规定 horizon 末端状态;
- 用于 recursive feasibility 和 closed-loop stability;
- 常见选择是 invariant terminal set。

11.4 Explicit MPC

Online MPC 每个 time step 求解 optimisation。Explicit MPC 先离线解 parametric optimisation, 把 state space 切成 regions:

- 每个 region 有一个对应的 affine feedback law;
- online 时只判断当前 state 落在哪个 region;
- 很快, 但 regions 数量可能指数增长。

适用场景:

- low-dimensional system;
- extremely high-speed control;
- embedded controller。

不适合:

- 高维系统;
- long horizon;
- constraints 很多导致 regions 爆炸。

12 Tutorial and Exercise Route

12.1 LS Exercises

重点题型：

1. BLUE proof

- unbiased condition: $Z^T \Phi = I$;
- covariance: $\text{Cov}(\hat{\theta}) = Z^T R Z$;
- BLUE estimator:

$$\hat{\theta}_Z = (\Phi^T R^{-1} \Phi)^{-1} \Phi^T R^{-1} Y$$

2. Polynomial and basis-function fitting

- 根据 basis functions 建立 design matrix;
- 判断 $X^T X$ 是否 nonsingular;
- radial basis functions 与 kernel matrix。

3. Conic section fitting

- 建立 parameter vector;
- 固定 scaling, 例如令 $f = 1$;
- 判断 ellipse/parabola/hyperbola。

12.2 Statistics Exercises

重点题型：

- true/false 判断:
 - regularised estimate 通常 biased;
 - inverse variance estimate unbiased;
 - inverse variance weighting 是 BLUE;
 - regularised estimate 可以有更低 MSE。
- 计算:
 - ordinary LS estimate;
 - covariance matrix;
 - non-identically distributed noise 下的 BLUE。

12.3 Convex Optimisation Exercises

重点题型：

- prove convexity:
 - $C = \{x \mid Ax \leq b\}$;

- intersection of convex sets;
- norm cone $\{(r, x) \mid \|x\| \leq r\}$.
- formulate as LP:
 - $\min \|Ax - b\|_\infty$;
 - $\min \|Ax - b\|_1$;
 - use epigraph variables and linear inequalities.

Lower priority:

- S-procedure minimum-volume ellipsoid problem is beyond examination detail, but the idea that ellipsoid containment can become an SDP is useful.

13 Exam Templates and Common Mistakes

13.1 Template: Form a Least-squares Problem

1. Identify output vector y .
2. Choose parameter vector θ .
3. Build each row of Φ from the model and data.
4. Write residual $r = y - \Phi\theta$.
5. Minimise $\|r\|_2^2$.
6. If full column rank, use $(\Phi^T\Phi)^{-1}\Phi^Ty$.

Common mistakes:

- using $\Phi\Phi^T$ instead of $\Phi^T\Phi$;
- forgetting the dimensions of θ ;
- treating noisy x and noisy y as ordinary LS without comment.

13.2 Template: Answer a BLUE Question

1. Start from $\hat{\theta} = Z^Ty$.
2. Use $y = \Phi\theta + \epsilon$.
3. For unbiasedness:

$$E[\hat{\theta}] = Z^T\Phi\theta = \theta \Rightarrow Z^T\Phi = I$$

4. Covariance:

$$\text{Cov}(\hat{\theta}) = Z^TRZ$$

5. State the BLUE:

$$\hat{\theta} = (\Phi^T R^{-1} \Phi)^{-1} \Phi^T R^{-1} y$$

Common mistakes:

- assuming $R = I$ when the question gives correlated noise;
- saying BLUE minimises MSE rather than variance among unbiased linear estimators.

13.3 Template: Explain Regularisation

Use this structure:

Regularisation adds a penalty term to the fitting objective.

It shrinks the estimate, which increases bias but decreases variance.

This can reduce MSE and improve prediction, especially when the problem is ill-conditioned.

The hyperparameter eta is usually selected by validation or cross-validation.

13.4 Template: Prove Convexity of a Set

1. Take arbitrary x_1, x_2 in the set.
2. Let $x_\gamma = \gamma x_1 + (1 - \gamma)x_2$, with $\gamma \in [0, 1]$.
3. Show x_γ also satisfies the set condition.
4. Conclude the set is convex.

For $C = \{x \mid Ax \leq b\}$:

$$A(\gamma x_1 + (1 - \gamma)x_2) = \gamma Ax_1 + (1 - \gamma)Ax_2 \leq \gamma b + (1 - \gamma)b = b$$

13.5 Template: Formulate MPC

1. State current state x_0 .
2. Define decision variables:
 - $u_0, \dots, u_{M-1}$;
 - $x_1, \dots, x_M$.
3. Write dynamics constraints.
4. Add hard constraints.
5. Add soft constraints as penalties if needed.
6. Add terminal cost or terminal constraint.
7. Solve, apply only u_0 , then replan.

Common mistakes:

- forgetting that only the first input is applied;
- treating forecasts as feedback;
- omitting state estimation even though state is usually estimated, not directly measured;
- writing terminal constraints without explaining their role.

14 Self Check

You should be able to answer these without notes:

1. What are the normal equations and where do they come from?
2. Why is the LS residual orthogonal to the column space of Φ ?
3. What is the difference between ordinary LS, weighted LS, and BLUE?
4. Why does minimum variance among unbiased estimators not imply minimum MSE?
5. How does regularisation change bias and variance?
6. What is a convex set?
7. Why are affine equality constraints allowed in convex optimisation?
8. What is the difference between LP, QP, SOCP, QCQP, and SDP?
9. What is an LMI?
10. What does the Schur complement help with?
11. How do you write an MPC optimisation problem?
12. Why does MPC apply only the first input?
13. What is explicit MPC, and why does it not scale well?

15 Minimum Memory List

- LS problem:

$$\min_{\theta} \|y - \Phi\theta\|_2^2$$

- Normal equations:

$$\Phi^T \Phi \theta = \Phi^T y$$

- Ordinary LS:

$$\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T y$$

- BLUE:

$$\hat{\theta} = (\Phi^T R^{-1} \Phi)^{-1} \Phi^T R^{-1} y$$

- MSE:

$$\text{MSE} = \text{bias}^2 + \text{variance}$$

- Regularised LS:

$$\hat{\theta} = (\Phi^T \Phi + \eta I)^{-1} \Phi^T y$$

- Convex set:

$$\gamma x_1 + (1 - \gamma) x_2 \in C$$

- PSD:

$$X \succeq 0 \Leftrightarrow z^T X z \geq 0 \quad \forall z$$

- MPC loop:

```
estimate state -> optimise horizon -> apply first input -> re-estimate -> repeat
```